

```
=====
N = 10 (h = 0.2000)
=====
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	t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd
RK4	Adams-Bashforth	Adams-Moulton	Pred-Corrector			
	0.00000000	0.50000000	0.50000000	0.50000000	0.50000000	0.50000000
	0.50000000	0.50000000	0.50000000		0.50000000	
	0.20000000	0.82929862	0.80000000	0.83000000	0.82800000	0.82924444
	0.82929333	0.82929333	0.82929333		0.82929333	
	0.40000000	1.21408765	1.15200000	1.21580000	1.21136000	1.21397499
	1.21407621	1.21407621	1.21407621		1.21407621	
	0.60000000	1.64894060	1.55040000	1.65207600	1.64465920	1.64876590
	1.64892202	1.64892202	1.64895060		1.64892202	
	0.80000000	2.12722954	1.98848000	2.13233272	2.12128422	2.12699053
	2.12720268	2.12728925	2.12727223		2.12720563	
	1.00000000	2.64085909	2.45817600	2.64864592	2.63316675	2.64055555
	2.64082269	2.64105333	2.64094815		2.64082860	
	1.20000000	3.17994154	2.94981120	3.19134802	3.17046344	3.17957629
	3.17989417	3.18031413	3.18009538		3.17990264	
	1.40000000	3.73240002	3.45177344	3.74864458	3.72116540	3.73198028
	3.73234007	3.73301859	3.73264296		3.73235048	
	1.60000000	4.28348379	3.95012813	4.30614639	4.27062178	4.28302303
	4.28340950	4.28444241	4.28384775		4.28342082	
	1.80000000	4.81517627	4.42815375	4.84629860	4.80095857	4.81469657
	4.81508569	4.81659556	4.81570293		4.81509636	
	2.00000000	5.30547195	4.86578450	5.34768429	5.29036946	5.30500719
	5.30536300	5.30750818	5.30621550		5.30537067	
	max error		0.43968745	0.04221234	0.01510249	0.00047969
	0.00010895	0.00203623	0.00074355		0.00010128	

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=====
N = 20 (h = 0.1000)
=====
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	t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd
RK4	Adams-Bashforth	Adams-Moulton	Pred-Corrector			
	0.00000000	0.50000000	0.50000000	0.50000000	0.50000000	0.50000000
	0.50000000	0.50000000	0.50000000		0.50000000	
	0.10000000	0.65741454	0.65000000	0.65750000	0.65725000	0.65741111
	0.65741437	0.65741437	0.65741437		0.65741437	
	0.20000000	0.82929862	0.81400000	0.82948750	0.82896125	0.82929162
	0.82929828	0.82929828	0.82929828		0.82929828	
	0.30000000	1.01507060	0.99140000	1.01538369	1.01455218	1.01505990
	1.01507006	1.01507006	1.01507087		1.01507006	
	0.40000000	1.21408765	1.18154000	1.21454897	1.21338016	1.21407315
	1.21408691	1.21408917	1.21408869		1.21408696	
	0.50000000	1.42563936	1.38369400	1.42627662	1.42473508	1.42562095
	1.42563840	1.42564366	1.42564132		1.42563850	
	0.60000000	1.64894060	1.59706340	1.64978566	1.64783226	1.64891820
	1.64893939	1.64894803	1.64894366		1.64893953	

0.70000000	1.88312365	1.82076974	1.88421316	1.88180465	1.88309721
1.88312218	1.88313484	1.88312802	1.88312233		
0.80000000	2.12722954	2.05384671	2.12860554	2.12569414	2.12719904
2.12722779	2.12724523	2.12723546	2.12722794		
0.90000000	2.38019844	2.29523139	2.38190912	2.37844202	2.38016392
2.38019640	2.38021947	2.38020620	2.38019653		
1.00000000	2.64085909	2.54375452	2.64295958	2.63887843	2.64082060
2.64085672	2.64088638	2.64086900	2.64085679		
1.10000000	2.90791699	2.79812998	2.91047033	2.90571067	2.90787468
2.90791428	2.90795164	2.90792942	2.90791426		
1.20000000	3.17994154	3.05694297	3.18301972	3.17751029	3.17989561
3.17993847	3.17998480	3.17995691	3.17993831		
1.30000000	3.45535167	3.31863727	3.45903679	3.45269887	3.45530241
3.45534821	3.45540495	3.45537046	3.45534787		
1.40000000	3.73240002	3.58150100	3.73678565	3.72953225	3.73234783
3.73239614	3.73246496	3.73242278	3.73239556		
1.50000000	4.00915546	3.84365110	4.01434814	4.00608313	4.00910085
4.00915115	4.00923394	4.00918282	4.00915026		
1.60000000	4.28348379	4.10301621	4.28960470	4.28022186	4.28342740
4.28347900	4.28357791	4.28351646	4.28347772		
1.70000000	4.55302630	4.35731783	4.56021319	4.54959516	4.55296896
4.55302101	4.55313850	4.55306510	4.55301925		
1.80000000	4.81517627	4.60404961	4.82358558	4.81160265	4.81511898
4.81517044	4.81530930	4.81522212	4.81516808		
1.90000000	5.06705278	4.84045457	5.07686206	5.06337093	5.06699677
5.06704639	5.06720979	5.06710674	5.06704331		
2.00000000	5.30547195	5.06350003	5.31688258	5.30172488	5.30541871
5.30546496	5.30565651	5.30553522	5.30546102		
max error		0.24197192	0.01141063	0.00374707	0.00005734
0.00000699	0.00018456	0.00006327	0.00001093		

=====

N = 40 (h = 0.0500)

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t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd
RK4 Adams-Bashforth Adams-Moulton Pred-Corrector					
0.00000000	0.50000000	0.50000000	0.50000000	0.50000000	0.50000000
0.50000000	0.50000000	0.50000000	0.50000000	0.50000000	
0.05000000	0.57686445	0.57500000	0.57687500	0.57684375	0.57686424
0.57686445	0.57686445	0.57686445	0.57686445	0.57686445	
0.10000000	0.65741454	0.65362500	0.65743672	0.65737262	0.65741411
0.65741453	0.65741453	0.65741453	0.65741453	0.65741453	
0.15000000	0.74158288	0.73580625	0.74161785	0.74151921	0.74158222
0.74158286	0.74158286	0.74158289	0.74158286	0.74158286	
0.20000000	0.82929862	0.82147156	0.82934764	0.82921270	0.82929773
0.82929860	0.82929866	0.82929865	0.82929860	0.82929860	
0.25000000	0.92048729	0.91054514	0.92055171	0.92037860	0.92048617
0.92048726	0.92048740	0.92048734	0.92048727	0.92048727	
0.30000000	1.01507060	1.00294740	1.01515186	1.01493863	1.01506924
1.01507056	1.01507078	1.01507067	1.01507057	1.01507057	

0.35000000	1.11296623	1.09859477	1.11306589	1.11281048	1.11296463
1.11296618	1.11296649	1.11296633	1.11296619		
0.40000000	1.21408765	1.19739951	1.21420739	1.21390764	1.21408581
1.21408760	1.21408801	1.21408778	1.21408761		
0.45000000	1.31834391	1.29926948	1.31848552	1.31813916	1.31834182
1.31834385	1.31834436	1.31834407	1.31834386		
0.50000000	1.42563936	1.40410796	1.42580478	1.42540942	1.42563703
1.42563930	1.42563992	1.42563957	1.42563931		
0.55000000	1.53587349	1.51181335	1.53606477	1.53561790	1.53587091
1.53587342	1.53587417	1.53587373	1.53587343		
0.60000000	1.64894060	1.62227902	1.64915997	1.64865894	1.64893777
1.64894052	1.64894140	1.64894088	1.64894053		
0.65000000	1.76472959	1.73539297	1.76497942	1.76442146	1.76472650
1.76472950	1.76473052	1.76472992	1.76472950		
0.70000000	1.88312365	1.85103762	1.88340649	1.88278869	1.88312031
1.88312355	1.88312473	1.88312403	1.88312355		
0.75000000	2.00399999	1.96908950	2.00431857	2.00363786	2.00399640
2.00399989	2.00400124	2.00400043	2.00399989		
0.80000000	2.12722954	2.08941898	2.12758677	2.12683992	2.12722569
2.12722942	2.12723096	2.12723003	2.12722942		
0.85000000	2.25267657	2.21188993	2.25307559	2.25225922	2.25267248
2.25267645	2.25267819	2.25267713	2.25267644		
0.90000000	2.38019844	2.33635942	2.38064259	2.37975313	2.38019410
2.38019831	2.38020027	2.38019907	2.38019830		
0.95000000	2.50964517	2.46267739	2.51013802	2.50917173	2.50964058
2.50964503	2.50964722	2.50964587	2.50964501		
1.00000000	2.64085909	2.59068626	2.64140447	2.64035741	2.64085425
2.64085894	2.64086138	2.64085987	2.64085891		
1.05000000	2.77367444	2.72022058	2.77427645	2.77314447	2.77366937
2.77367428	2.77367699	2.77367531	2.77367425		
1.10000000	2.90791699	2.85110660	2.90858000	2.90735875	2.90791168
2.90791682	2.90791982	2.90791795	2.90791678		
1.15000000	3.04340355	2.98316193	3.04413222	3.04281714	3.04339802
3.04340336	3.04340668	3.04340461	3.04340331		
1.20000000	3.17994154	3.11619503	3.18074087	3.17932714	3.17993580
3.17994134	3.17994501	3.17994272	3.17994129		
1.25000000	3.31732852	3.25000478	3.31820384	3.31668641	3.31732257
3.31732831	3.31733234	3.31732982	3.31732825		
1.30000000	3.45535167	3.38438002	3.45630866	3.45468221	3.45534552
3.45535145	3.45535587	3.45535309	3.45535136		
1.35000000	3.59378723	3.51909902	3.59483198	3.59309092	3.59378091
3.59378700	3.59379185	3.59378879	3.59378691		
1.40000000	3.73240002	3.65392897	3.73353900	3.73167746	3.73239353
3.73239977	3.73240507	3.73240172	3.73239966		
1.45000000	3.87094274	3.78862542	3.87218287	3.87019468	3.87093611
3.87094248	3.87094827	3.87094461	3.87094235		
1.50000000	4.00915546	3.92293169	4.01050412	4.00838278	4.00914871
4.00915519	4.00916150	4.00915750	4.00915504		
1.55000000	4.14676491	4.05657828	4.14822995	4.14596865	4.14675805
4.14676462	4.14677149	4.14676713	4.14676445		

1.60000000	4.28348379	4.18928219	4.28507361	4.28266517	4.28347685
4.28348348	4.28349096	4.28348620	4.28348329		
1.65000000	4.41901009	4.32074630	4.42073364	4.41817051	4.41900309
4.41900977	4.41901789	4.41901271	4.41900955		
1.70000000	4.55302630	4.45065862	4.55489311	4.55216737	4.55301929
4.55302597	4.55303478	4.55302915	4.55302572		
1.75000000	4.68519866	4.57869155	4.68721888	4.68432220	4.68519166
4.68519831	4.68520786	4.68520175	4.68519804		
1.80000000	4.81517627	4.70450113	4.81736072	4.81428433	4.81516932
4.81517590	4.81518625	4.81517961	4.81517559		
1.85000000	4.94259024	4.82772618	4.94495046	4.94168516	4.94258338
4.94258985	4.94260105	4.94259386	4.94258951		
1.90000000	5.06705278	4.94798749	5.06960105	5.06613715	5.06704606
5.06705237	5.06706448	5.06705670	5.06705199		
1.95000000	5.18815621	5.06488687	5.19090560	5.18723292	5.18814968
5.18815579	5.18816887	5.18816045	5.18815536		
2.00000000	5.30547195	5.17800621	5.30843639	5.30454424	5.30546566
5.30547151	5.30548563	5.30547653	5.30547104		
max error		0.12746574	0.00296444	0.00092771	0.00000701
0.00000044	0.00001368	0.00000458	0.00000091		

Error table | h = 0.2 (N = 10)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	0.500000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00	0.00e+00				
0.20000000	0.829299	2.93e-02	7.01e-04	1.30e-03	5.42e-05	5.29e-06
5.29e-06	5.29e-06	5.29e-06				
0.40000000	1.214088	6.21e-02	1.71e-03	2.73e-03	1.13e-04	1.14e-05
1.14e-05	1.14e-05	1.14e-05				
0.60000000	1.648941	9.85e-02	3.14e-03	4.28e-03	1.75e-04	1.86e-05
1.86e-05	1.00e-05	1.86e-05				
0.80000000	2.127230	1.39e-01	5.10e-03	5.95e-03	2.39e-04	2.69e-05
5.97e-05	4.27e-05	2.39e-05				
1.00000000	2.640859	1.83e-01	7.79e-03	7.69e-03	3.04e-04	3.64e-05
1.94e-04	8.91e-05	3.05e-05				
1.20000000	3.179942	2.30e-01	1.14e-02	9.48e-03	3.65e-04	4.74e-05
3.73e-04	1.54e-04	3.89e-05				
1.40000000	3.732400	2.81e-01	1.62e-02	1.12e-02	4.20e-04	5.99e-05
6.19e-04	2.43e-04	4.95e-05				
1.60000000	4.283484	3.33e-01	2.27e-02	1.29e-02	4.61e-04	7.43e-05
9.59e-04	3.64e-04	6.30e-05				
1.80000000	4.815176	3.87e-01	3.11e-02	1.42e-02	4.80e-04	9.06e-05
1.42e-03	5.27e-04	7.99e-05				
2.00000000	5.305472	4.40e-01	4.22e-02	1.51e-02	4.65e-04	1.09e-04
2.04e-03	7.44e-04	1.01e-04				

Error table | $h = 0.1$ ($N = 20$)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	0.500000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00		0.00e+00			
0.10000000	0.657415	7.41e-03	8.55e-05	1.65e-04	3.43e-06	1.66e-07
1.66e-07	1.66e-07		1.66e-07			
0.20000000	0.829299	1.53e-02	1.89e-04	3.37e-04	7.00e-06	3.45e-07
3.45e-07	3.45e-07		3.45e-07			
0.30000000	1.015071	2.37e-02	3.13e-04	5.18e-04	1.07e-05	5.38e-07
5.38e-07	2.72e-07		5.38e-07			
0.40000000	1.214088	3.25e-02	4.61e-04	7.07e-04	1.45e-05	7.45e-07
1.52e-06	1.04e-06		6.90e-07			
0.50000000	1.425639	4.19e-02	6.37e-04	9.04e-04	1.84e-05	9.69e-07
4.30e-06	1.96e-06		8.67e-07			
0.60000000	1.648941	5.19e-02	8.45e-04	1.11e-03	2.24e-05	1.21e-06
7.43e-06	3.06e-06		1.07e-06			
0.70000000	1.883124	6.24e-02	1.09e-03	1.32e-03	2.64e-05	1.47e-06
1.12e-05	4.37e-06		1.32e-06			
0.80000000	2.127230	7.34e-02	1.38e-03	1.54e-03	3.05e-05	1.75e-06
1.57e-05	5.93e-06		1.60e-06			
0.90000000	2.380198	8.50e-02	1.71e-03	1.76e-03	3.45e-05	2.04e-06
2.10e-05	7.76e-06		1.92e-06			
1.00000000	2.640859	9.71e-02	2.10e-03	1.98e-03	3.85e-05	2.36e-06
2.73e-05	9.91e-06		2.29e-06			
1.10000000	2.907917	1.10e-01	2.55e-03	2.21e-03	4.23e-05	2.70e-06
3.47e-05	1.24e-05		2.73e-06			
1.20000000	3.179942	1.23e-01	3.08e-03	2.43e-03	4.59e-05	3.07e-06
4.33e-05	1.54e-05		3.22e-06			
1.30000000	3.455352	1.37e-01	3.69e-03	2.65e-03	4.93e-05	3.46e-06
5.33e-05	1.88e-05		3.80e-06			
1.40000000	3.732400	1.51e-01	4.39e-03	2.87e-03	5.22e-05	3.88e-06
6.49e-05	2.28e-05		4.45e-06			
1.50000000	4.009155	1.66e-01	5.19e-03	3.07e-03	5.46e-05	4.32e-06
7.85e-05	2.74e-05		5.21e-06			
1.60000000	4.283484	1.80e-01	6.12e-03	3.26e-03	5.64e-05	4.79e-06
9.41e-05	3.27e-05		6.07e-06			
1.70000000	4.553026	1.96e-01	7.19e-03	3.43e-03	5.73e-05	5.29e-06
1.12e-04	3.88e-05		7.06e-06			
1.80000000	4.815176	2.11e-01	8.41e-03	3.57e-03	5.73e-05	5.83e-06
1.33e-04	4.59e-05		8.18e-06			
1.90000000	5.067053	2.27e-01	9.81e-03	3.68e-03	5.60e-05	6.39e-06
1.57e-04	5.40e-05		9.47e-06			
2.00000000	5.305472	2.42e-01	1.14e-02	3.75e-03	5.32e-05	6.99e-06
1.85e-04	6.33e-05		1.09e-05			

Error table | $h = 0.05$ ($N = 40$)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	0.500000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00		0.00e+00			
0.05000000	0.576864	1.86e-03	1.05e-05	2.07e-05	2.16e-07	5.20e-09
5.20e-09	5.20e-09		5.20e-09			
0.10000000	0.657415	3.79e-03	2.22e-05	4.19e-05	4.36e-07	1.06e-08
1.06e-08	1.06e-08		1.06e-08			
0.15000000	0.741583	5.78e-03	3.50e-05	6.37e-05	6.60e-07	1.62e-08
1.62e-08	7.94e-09		1.62e-08			
0.20000000	0.829299	7.83e-03	4.90e-05	8.59e-05	8.88e-07	2.20e-08
4.29e-08	2.86e-08		2.07e-08			
0.25000000	0.920487	9.94e-03	6.44e-05	1.09e-04	1.12e-06	2.80e-08
1.12e-07	5.14e-08		2.55e-08			
0.30000000	1.015071	1.21e-02	8.13e-05	1.32e-04	1.36e-06	3.43e-08
1.85e-07	7.63e-08		3.08e-08			
0.35000000	1.112966	1.44e-02	9.97e-05	1.56e-04	1.60e-06	4.08e-08
2.66e-07	1.04e-07		3.66e-08			
0.40000000	1.214088	1.67e-02	1.20e-04	1.80e-04	1.84e-06	4.75e-08
3.55e-07	1.34e-07		4.29e-08			
0.45000000	1.318344	1.91e-02	1.42e-04	2.05e-04	2.08e-06	5.45e-08
4.52e-07	1.67e-07		4.98e-08			
0.50000000	1.425639	2.15e-02	1.65e-04	2.30e-04	2.33e-06	6.18e-08
5.58e-07	2.02e-07		5.72e-08			
0.55000000	1.535873	2.41e-02	1.91e-04	2.56e-04	2.58e-06	6.93e-08
6.74e-07	2.41e-07		6.53e-08			
0.60000000	1.648941	2.67e-02	2.19e-04	2.82e-04	2.83e-06	7.71e-08
8.01e-07	2.84e-07		7.40e-08			
0.65000000	1.764730	2.93e-02	2.50e-04	3.08e-04	3.08e-06	8.52e-08
9.39e-07	3.30e-07		8.35e-08			
0.70000000	1.883124	3.21e-02	2.83e-04	3.35e-04	3.34e-06	9.35e-08
1.09e-06	3.81e-07		9.37e-08			
0.75000000	2.004000	3.49e-02	3.19e-04	3.62e-04	3.59e-06	1.02e-07
1.25e-06	4.35e-07		1.05e-07			
0.80000000	2.127230	3.78e-02	3.57e-04	3.90e-04	3.84e-06	1.11e-07
1.43e-06	4.94e-07		1.17e-07			
0.85000000	2.252677	4.08e-02	3.99e-04	4.17e-04	4.10e-06	1.20e-07
1.62e-06	5.58e-07		1.30e-07			
0.90000000	2.380198	4.38e-02	4.44e-04	4.45e-04	4.35e-06	1.30e-07
1.83e-06	6.28e-07		1.44e-07			
0.95000000	2.509645	4.70e-02	4.93e-04	4.73e-04	4.59e-06	1.40e-07
2.05e-06	7.03e-07		1.59e-07			
1.00000000	2.640859	5.02e-02	5.45e-04	5.02e-04	4.84e-06	1.50e-07
2.29e-06	7.84e-07		1.75e-07			
1.05000000	2.773674	5.35e-02	6.02e-04	5.30e-04	5.07e-06	1.61e-07
2.55e-06	8.71e-07		1.92e-07			
1.10000000	2.907917	5.68e-02	6.63e-04	5.58e-04	5.30e-06	1.72e-07
2.83e-06	9.66e-07		2.11e-07			
1.15000000	3.043404	6.02e-02	7.29e-04	5.86e-04	5.53e-06	1.83e-07
3.14e-06	1.07e-06		2.31e-07			

1.20000000	3.179942	6.37e-02	7.99e-04	6.14e-04	5.74e-06	1.95e-07
3.47e-06	1.18e-06		2.53e-07			
1.25000000	3.317329	6.73e-02	8.75e-04	6.42e-04	5.95e-06	2.07e-07
3.82e-06	1.30e-06		2.76e-07			
1.30000000	3.455352	7.10e-02	9.57e-04	6.69e-04	6.14e-06	2.20e-07
4.20e-06	1.42e-06		3.01e-07			
1.35000000	3.593787	7.47e-02	1.04e-03	6.96e-04	6.32e-06	2.33e-07
4.61e-06	1.56e-06		3.28e-07			
1.40000000	3.732400	7.85e-02	1.14e-03	7.23e-04	6.49e-06	2.46e-07
5.05e-06	1.71e-06		3.57e-07			
1.45000000	3.870943	8.23e-02	1.24e-03	7.48e-04	6.63e-06	2.60e-07
5.53e-06	1.86e-06		3.88e-07			
1.50000000	4.009155	8.62e-02	1.35e-03	7.73e-04	6.76e-06	2.74e-07
6.04e-06	2.03e-06		4.21e-07			
1.55000000	4.146765	9.02e-02	1.47e-03	7.96e-04	6.86e-06	2.89e-07
6.58e-06	2.22e-06		4.57e-07			
1.60000000	4.283484	9.42e-02	1.59e-03	8.19e-04	6.94e-06	3.04e-07
7.17e-06	2.41e-06		4.95e-07			
1.65000000	4.419010	9.83e-02	1.72e-03	8.40e-04	6.99e-06	3.20e-07
7.80e-06	2.62e-06		5.36e-07			
1.70000000	4.553026	1.02e-01	1.87e-03	8.59e-04	7.01e-06	3.36e-07
8.48e-06	2.85e-06		5.79e-07			
1.75000000	4.685199	1.07e-01	2.02e-03	8.76e-04	7.00e-06	3.52e-07
9.20e-06	3.09e-06		6.26e-07			
1.80000000	4.815176	1.11e-01	2.18e-03	8.92e-04	6.95e-06	3.69e-07
9.98e-06	3.35e-06		6.76e-07			
1.85000000	4.942590	1.15e-01	2.36e-03	9.05e-04	6.86e-06	3.87e-07
1.08e-05	3.62e-06		7.29e-07			
1.90000000	5.067053	1.19e-01	2.55e-03	9.16e-04	6.72e-06	4.05e-07
1.17e-05	3.92e-06		7.87e-07			
1.95000000	5.188156	1.23e-01	2.75e-03	9.23e-04	6.53e-06	4.23e-07
1.27e-05	4.24e-06		8.48e-07			
2.00000000	5.305472	1.27e-01	2.96e-03	9.28e-04	6.29e-06	4.42e-07
1.37e-05	4.58e-06		9.13e-07			